

A NOVEL END-TO-END SPEECH EMOTION RECOGNITION NETWORK WITH STACKED TRANSFORMER LAYERS

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ABSTRACT

Speech emotion recognition (SER) aims to automatically recognize emotional category for a given speech utterance. The performance of a SER system heavily relies on the effectiveness of global representation expressed at utterance level. To effectively extract such a global feature, the mainstream of recent SER architectures adopts a pipeline with two key modules, feature extraction and aggregation. Although variant module designs have brought impressive progresses, SER is still a challenging task. In contrast with those previous works, herein we propose a novel strategy for global SER feature extraction by applying an additional enhancement module on top of the current SER pipeline. To verify its effect, an end-to-end SER architecture is proposed where stacked multiple transformer layers are explored to enhance the aggregated global feature. Such an architecture is evaluated on IEMOCAP and results strongly substantiate the effectiveness of our proposal. In terms of weighted accuracy on four emotion categories, our proposed SER system outperforms the prior arts by a large margin of relatively 20% improvement. Our codes and the pre-trained SER models are made publicly available.

Index Terms— Speech emotion recognition, end-to-end, stacked transformer layers

1. INTRODUCTION

Speech emotion, as one kind of meta-information apart from text, plays an important role for understanding speakers' psychology and response. The relevant research, called speech emotion recognition (SER), aims to automatically recognize emotional category for a given speech utterance. Since emotions are usually conveyed in a subtle and variable way, it has been challenging to identify emotion embeddings, as representation of an utterance, that can effectively classify emotion categories.

With recent advances in Deep Neural Networks (DNNs), emotional embeddings have evolved from prior knowledge-based hand-crafted acoustic features, e.g. low-level descriptors (LLDs), to DNN-based deep emotion features. Various DNN architectures, like convolutional neural network (CNN), long-short term memory (LSTM), time-delay neural network

(TDNN), residual network (ResNet), Dilated Residual Network (DRN), themselves or in combinations, have been explored in most recent SER works [1–4].

A lot of research works have been afforded in the SER area and it is still challenging because of decoupling emotion from linguistic feature and pooling valid emotion features from a long span of utterance. To address these, it became prevalent for a modern SER system to adopt a pipeline with two-modules: 1) a feature extraction module to generate temporal acoustic features that are emotionally relevant; and 2) an aggregation module to pool those temporal features into a compact global contextual representation (aka, emotional embedding) at utterance level.

To yield effective emotional embeddings, recent SER works have been focused on developing variant module architectures. For example, [2] proposed a mixed CNN-LSTM architecture, where CNN extracts feature sequences from raw spectrograms, and LSTM aggregates long-term feature dependencies. Similarly, both 1D and 2D CNN-LSTM based SER system were proposed in [5] and showed that 2D CNN-LSTM network outperformed the 1D one by capturing both local correlations and global contextual information from spectrograms. Later motivated by attention mechanism [6], RNN-attention was proposed in [7] where RNN extracts temporal features and attention aggregates long-term feature dependencies by focusing on emotionally salient features. A few temporal modeling approaches were tested in [8], aiming to learn deep emotion features from raw waveform. And authors in [9] introduced CNN-BLSTM-attention architecture where attention was extended to multi-head attention (MHA), to explore different representation subspaces at different positions. The mechanism of MHA was again adopted by [4] to build a DRN-attention architecture instead, where DRN enables the network to retain high resolution of temporal structure in feature learning. Further, contextual LSTM-attention was proposed in [10] to consider the relation and dependencies among the surrounding utterances.

Notwithstanding the progresses made in previous works, current SER performances still leave room for improvement. For example, the SOTA classification accuracy on a benchmark dataset, IEMOCAP, is only around 76% (on four emotion categories). Meanwhile, we noted that in other

research domains, it has been frequently reported that replacing recurrence with stacked transformer layers (STLs) can yield significant performance improvements. For example, STLs-based acoustic models [11] gave the best acoustic models on Librispeech benchmark. In question answering domain, both stacked latent attention and multihop attention networks (MAN) [12] exhibited substantial performance improvements. For image captioning task, state-of-the-art (SOTA) results were achieved from stacked cross attention network (SCAN) [13] or stacked attention modules [14].

Inspired by those successes in other research fields, in this study, we are interested in leveraging the STL mechanism into SER network. In particular, a new SER architecture built by adding STLs on top of the current SER pipeline, is proposed. The motivation behind our proposal is to use strong feature learning ability of STLs module to directly enhance the pipeline output. To the authors' best knowledge, this is the first time in the literature to study SER problem using this strategy.

This paper is organized as follows. In Section 2, vanilla Transformer is briefly introduced. In Section 3, we propose STL-enhanced SER architecture in details. Experiments are reported in Section 4. And section 5 concludes the study.

2. BACKGROUND - STACKED TRANSFORMER LAYERS IN VANILLA TRANSFORMER

In this section, we briefly describe details about STLs that originated from vanilla Transformer [15], which relies entirely on attention mechanisms to capture long-range dependencies between input and output sequence. In this study, we are interested in the its encoder that transforms an input sequence into a global context vector with fixed dimension.

In detail, the encoder is built with as a stack of six identical transformer layers (TLs). Each TL contains two sub-layers with residual connections: 1) a self-attention sub-layer and 2) a position-wise feed-forward network (FFN) sub-layer. In addition, positional encoding (PE) is introduced to provide explicit order information to the model.

Benefited from the self-attention mechanism, the output sequence retains and leverages the input information by placing importance on specific parts of the input. TLs are stacked in way that only the first layer takes input sequence, every other layer just takes the preceding output as inputs. This enables contextual vector to be enriched via iterative attention and projection within latent representation space.

Mathematically, Transformer architecture is interpreted [16] as a numerical ordinary differential equation (ODE) solver for a convection-diffusion equation in a multi-particle dynamic system. From this viewpoint, the number of STLs corresponds to the time dimension in ODE, which naturally empowers Transformer with strong learning ability to produce deep contextual representation.

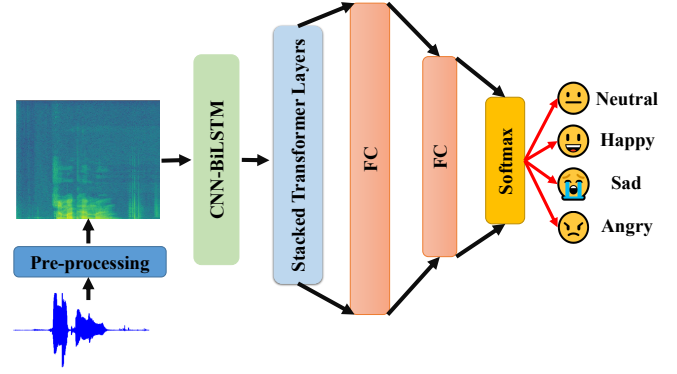


Fig. 1. Our proposed SER architecture

3. PROPOSED METHOD

Our proposed new SER framework is introduced in this section. Fig.1 gives an overview of its architecture. As illustrated, it has 4 modules in sequence: front-end pre-processing module to extract frame-level LLDs, CNN-BiLSTM module to extract contextual representation, STLs to enhance the contextual representation and back-end classifier to predict probability of emotion categories. Design details of both CNN-BiLSTM and STLs module are introduced below.

3.1. CNN-BiLSTM Module

Details of our CNN-BiLSTM module is shown in Fig.2, which resembles our previous study [17]. To recap, this module is constructed with six pairs of CNN-pooling layers and one BiLSTM layer. Each CNN-pooling pair includes one convolution layer (equipped with batch normalization and activated by Elu function) to extract the high-level temporal features, and one pooling layer (with fixed dropout rate) to reduce the feature dimension.

In particular, the module input is frame-level based LLD acoustic features, presented as $X = \{x_1, x_2, \dots, x_t\} \in \mathbb{R}^{t \times d}$, where t denote the frame length and d denotes the LLD feature dimension. For the 1st CNN-pooling pair, 2D-convolutions, with ch_0 filters, are applied over X and produces a temporal feature sequence, expressed as $H_0^{cnn} = \{h_1^{cnn}, h_2^{cnn}, \dots, h_t^{cnn}\} \in \mathbb{R}^{t \times h_0 \times w_0}$; then 2D-average-pooling operation is applied to every $n_{ap} * n_{ap}$ CNN outputs to reduce dimension. The pooled output $H_0^{pool} = \{h_1^{pool}, h_2^{pool}, \dots, h_{ch_0}^{pool}\} \in \mathbb{R}^{ch_0 \times h_0/n_{ap} \times w_0/n_{ap}}$ is sequentially fed to the next pair. By repeating same operations on all 6 CNN-pooling pairs, we obtain the final latent representation $H_5^{pool} \in \mathbb{R}^{ch_5 \times h_5 \times w_5}$. With further reshape operation and its reshaped version $H^{re} \in \mathbb{R}^{h_5 \times (ch_5 \times w_5)}$ subsequently feeds to the BiLSTM layer.

BiLSTM learns contextual dependencies from H^{re} both forward and backward. Its output, a hidden state sequence $H^{bilstm} = \{h_1^{bilstm}, h_2^{bilstm}, \dots, h_{h_5}^{bilstm}\}$, represents a contextual feature sequence, where $h_i^{bilstm} = [\vec{h}^{bilstm}, \overleftarrow{h}^{bilstm}] \in \mathbb{R}^{2d_{lstm}}$ with the size of the hidden state as d_{lstm} .

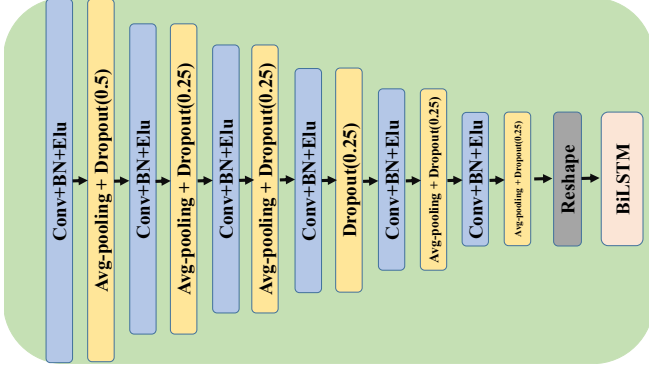


Fig. 2. Detailed architecture of CNN-BiLSTM module in Fig.1.

In classic SER pipeline, two kinds of aggregation approaches, namely recurrence-based and attention-based, are applied to yield global emotional embedding.

$$X^G = \begin{cases} h_{h_5}^{\text{bilstm}}, & \text{recurrence - based,} \\ \sum_{i=1} h_5 \alpha_i h_i^{\text{bilstm}}, & \text{attention - based,} \end{cases} \quad (1)$$

where α_i denotes attention weight.

In either way, the emotional embedding X^G (or subject to further affine transformation), is fed into the back-end classifier for probability prediction.

3.2. Stacked Transformer Layers

As our core contribution, an alternative approach is proposed to produce emotional embedding. It applies additional STLs to enhance the said contextual feature sequence H^{bilstm} .

To this end, 1D-convolution is performed first to map H^{bilstm} to vector $Z_0 \in \mathbb{R}^{2d_{\text{lstsm}}}$. Then STLs with L layers are employed to iteratively enhance Z_0 as: $Z_i = G(Z_{i-1})$ and. A series of operations are performance within G , such as multi-head attention, feed-forward and residual connection (like described in Section 2), except that positional encoding is skipped. The output of last Transformer layer is the global embedding, i.e. $X^G = Z_{L-1}$.

Lastly, the proposed SER architecture, as a whole, is trained in an end-to-end manner with the objective function as follows:

$$\hat{y}_c = \text{softmax}(X_G^T W + b), \quad (2)$$

$$L = -\log \prod_{i \in S} \sum_{c=1}^C \hat{y}_{i,c} \log(\hat{y}_{i,c}), \quad (3)$$

where S denotes the sample set for training and C is the total number of emotion categories.

We note that a similar architecture is leveraged for multimodal emotion recognition [18]. Our approach is different from the earlier work, as CMA is operated on temporal feature sequence which needs further statistics pooling; while

in our case, STLs apply on utterance-level representation instead.

Regarding detailed architecture parameters, please refer to our code repository ¹.

4. EXPERIMENTS

For performance evaluation, we train and test the proposed SER system on IEMOCAP [19], a benchmark dataset widely used for SER research. Experimental details and results are presented in this section.

4.1. Experiment Setup

Datasets. IEMOCAP contains more than 10K utterances, annotated by labels of nine emotion categories. Following reported procedure in prior works, a four-class ($\{happy, angry, sad, neutral\}$) sub-database, with total 5531 utterances, is used in our experiments. For each emotion class, its associated samples are randomly split into train/val/test in ratio of 7/1/2, respectively.

Evaluation Metrics. Both weighted accuracy (WA) and unweighted accuracy (UA) are employed for performances assessment.

Implementation Details. The proposed SER system is implemented in PyTorch. Adam [20] optimizer is used to optimize categorical cross entropy, meanwhile validation accuracy is monitored with early stopping setting as 8 epochs. The system is trained with batch size of 32, learning rate of 2×10^{-4} and decay rate of 1×10^{-6} .

4.2. Evaluation Results Compared with Prior Works

Regarding the pre-processing, for each utterance, firstly its duration is clipped into 7.5 seconds by trimming or zero padding, then a40-dimensional log Mel-filterbank energy features are extracted on frame basis (with window length of 25 and hop size of 10 ms). To prevent overfitting, 5-fold cross-validation technique is adopted in our experiments.

The final system performance is listed in Table 1 and compared with top-performing SER models.

Note that dataset splitting choices in these works are, in fact, not rigorously consistent, which makes it difficult for a fair comparison. As such, only performance trend could be observed by citing those originally reported best performances. A quick observation shows the WA metric has been consistently improved over time, ranging from 63.5% to 84%. However, we clarify that there are 3 cases (highlighted with footnotes) that adopted different input/output from the others. In particular, [2] used augmented input; [9] adopted multi-task loss requiring gender information; and [21] used cleaned spectrum as input. In view of these, we claim that existing

¹https://github.com/HW-AARC-CLUB/ICASSP_SER

SOTA WA on IEMOCAP is 76%, in the context of same network I/O.

Benefited from the new architecture, our SER system achieved significant improvement on the IEMOCAP dataset. With WA of 92.8%, it outperformed all existing works by a substantial margin of 20% relative improvement.

Table 1. Comparison with the recent SOTAs on IEMOCAP dataset.

Model	Year	WA	UA
RNN+Attn [7]	2017	63.5%	58.8%
CNN + LSTM [2]	2018	66.9% *	65.3% *
TDNN-LSTM-Attn [8]	2018	70.1%	60.7%
DRN + MHA [4]	2019	-	67.4%
IAAN [22]	2019	64.7%	66.3%
MHA [9]	2019	70.5% 81.6% **	72.6% 82.8% **
DSCNN [21]	2019	76% 84% ***	72% 82% ***
Ours	2020	91.28%	92%

* with data-augmentation

** with multi-task learning application

*** with clean spectrogram free of silences and noises

The classification results on individual category are further reported in Table 2. As expected, *neutral* is the most difficult category to predict, coinciding with observations reported in previous works [17, 22, 23].

Table 2. Confusion matrix of our SER system

Labels	Predictions			
	Neutral	Happy	Sad	Angry
Neutral	1479	106	119	4
Happy	109	1484	3	40
Sad	43	30	1011	0
Angry	24	0	2	1077

4.3. Further Investigations on Effectiveness of STLs

More investigations are conducted to assess the effectiveness of our proposed STL module. Firstly, a truncated SER system without the STLs module is investigated. The resulting WA plunges to 72.79%. The meaning of this investigation is two-fold: 1) our truncated system is well-designed with promising performance comparable to prior arts; 2) the STLs module provides massive contribution (18.5%) to the overall system performance, at the cost of reasonable increase in number of parameters (from 356K to 468k).

Secondly, the effectiveness of STLs is checked by visualizing the feature quality discovered by the intermediate layer using t-SNE technique [24]. Two latent feature distributions, Z_0 and Z_4 , are compared and illustrated in Fig.3 with 2D plots regarding the classes. Again, it can be observed that features after STLs are clearly projected to different clusters and well separated from each other.

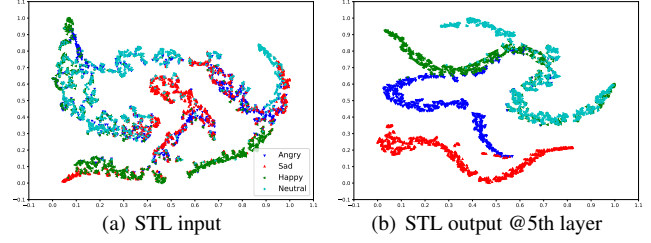


Fig. 3. Visualization for distributions of global contextual feature.

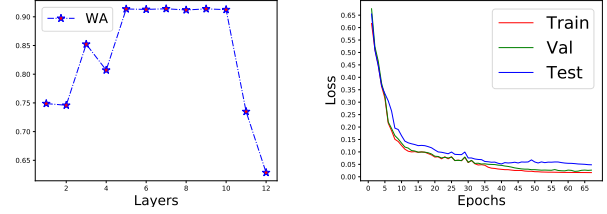


Fig. 4. Further investigation results on STLs

4.4. Effect of STL Depth

Lastly, the optimal layer number of STLs further investigated. The experimental results are illustrated in Fig.4(a), which shows an interesting observation that the system performances are boosted quickly at the beginning (< 5 layers), and kept nearly flat in the middle (5-10 layers) then dropped dramatically in the end (> 10 layers). Based on this observation, the optimal depth for our proposed STLs module is chosen as 5.

Meanwhile, according learning curves of our model, with optimal 5 Transformer layers, are plotted in Fig.4(b), where the best model is achieved at the 67th epoch.

5. CONCLUSION

In this study, we proposed a new end-to-end SER architecture with novelty of placing additional stacked transformer layers (STLs) on the top of a typical SER architecture. The motivation behind is that STLs might further improve the representation ability. Experimental results on IEMOCAP validated our hypothesis by showing up to 18.5% boost in performance when STLs is introduced. Furthermore, the overall system outperformed the current SOTA with significant 20% relative improvement. In our future work the generalizability of our proposal would be investigated on a range of public datasets.

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